Interpolări

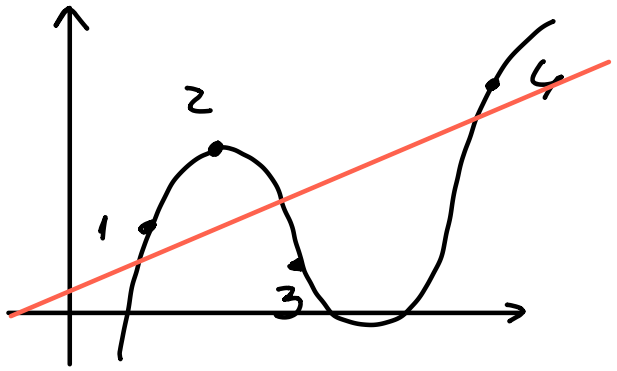
$x_0 \quad x_1 \ldots \ldots x_n$ → suportul interpolării

$y_0 \quad y_1 \ldots \ldots y_n$

→ se încearcă să se găsească valori intermediare între două valori date

$f: [x_0, x_n] \longrightarrow \mathbb{R}$

(vrem să aflăm funcția asociată punctelor de mai sus ca să le aproximez)



⇒ Vrem un polinom / o funcție a.î.

$P(x_i) = y_i$ sau $f(x_i) = y_i$

Polinom de interpolare

$$P_n(x) = \sum_{i=0}^{n} a_i u_i(x)$$

$u_0, u_1, \ldots u_n$ – baza

! baza are elemente diferite

$P_n(x_k) = y_k \quad \forall k = 0 : n$ → Condiția de interpolare

Th Weistrass

$\forall f$ o functie continua cunoscuta intr-un set de puncte x_k $\exists$ un polinom $P_n(x)$ numit polinom de interpolare ce $P_n(x_k) = y_k$ astfel incat $|f(x) - P_n(x)| < \varepsilon$

Metoda FFT

$$142 = 1 \cdot 10^2 + 4 \cdot 10 + 2 \cdot 1$$
$$= x^2 + 4x + 2$$

Atunci avem convolutia polinoamelor

$$P = P_a \cdot P_b = (a_n x^n + a_{n-1} x^{n-1} + \dots + a_0)(b_n x^n + \dots + b_0)$$

atunci x din P are coef $x^k \to a_0 b_k + a_1 b_{k-1} + \dots + a_k b_0$

$$P = P_a \cdot P_b \Rightarrow P = \sum c_k x^k$$

Alg $x_1, x_2, \dots x_n$ si $P_a = [P_a(x_1) \dots P_a(x_n)]$
$P_b = [P_b(x_1) \dots P_b(x_n)]$

atunci $P = [P_a(x_1) P_b(x_1) \dots \dots P_a(x_n) P_b(x_n)]$

Metodele de aflare P sunt:

(1) Vandermonde

$$\begin{cases} P(x_1) = y_1 \\ \vdots \\ P(x_n) = y_n \end{cases} = \begin{cases} a_0 + a_1 x_1 + a_2 x_1^2 + \dots + a_n x_1^n = y_1 \\ \vdots \\ a_0 + a_1 x_n + a_2 x_n^2 + \dots + a_n x_n^n = y_n \end{cases}$$

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^n \\ \vdots & & & & \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Metodă proastă pt. că
- → la valori mici ($x \to 0$) $\Rightarrow x^n = 0$
- → oscilații
- → instabil numerică

(2) Polinomul Lagrange

Idee $u_i(x_k) \overset{not}{=\!=} u_{ik}$

$$U = (u_{ik})_{0 \le i,k \le n}$$

$$a = \begin{bmatrix} a_0 \\ \vdots \\ a_n \end{bmatrix} \qquad f = \begin{bmatrix} y_0 \\ \vdots \\ y_n \end{bmatrix} \qquad y_k = f(x_k)$$

$$\sum_{i=0}^{n} a_i u_{ik} = y_k \qquad k = 0:n$$

$$U \cdot a = f \Rightarrow a = U^{-1} \cdot f$$

$$P_n(x) = \sum_{i=0}^{n} a_i u_i(x) = a^T \cdot u(x) \qquad u(x) = \begin{bmatrix} u_0(x) \\ \vdots \\ u_n(x) \end{bmatrix}$$

$$= (U^{-1} \cdot f)^T \cdot u(x) = f^T \cdot \underbrace{(U^{-1})^T \cdot u(x)}_{l(x)}$$

$$= f^T \cdot l(x) = \sum_{i=0}^{n} y_i \cdot l_i(x)$$

$$P_n(x_k) = y_k \qquad \sum_{i=0}^{n} y_i l_i(x_k) = y_k$$

$$l_i(x_k) = \begin{cases} 0, & i \neq k \\ 1, & i = k \end{cases}$$

$$l_i(x) = C_i (x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)$$

$$= C_i \prod_{\substack{k=0 \\ k \neq i}}^{n} (x - x_k)$$

$$C_i = \frac{1}{\prod_{\substack{k=0 \\ k \neq i}}^{n} (x_i - x_k)} \Longrightarrow l_i(x) = \prod_{\substack{k=0 \\ k \neq i}}^{n} \frac{x - x_k}{x_i - x_k}$$

Multiplicatore Lagrange

$$P_n(x) = \sum_{i=1}^{n} y_i \cdot \prod_{\substack{k=0 \\ k \neq i}}^{n} \frac{x - x_k}{x_i - x_k}$$

Explicatie

$L_i(x_k) = 0$ ce $k \neq i$ pt. că

$$L_i(x_k) = \frac{(x-x_1)\dots(x-x_{k-1})(x-x_k)\dots}{\dots} \Rightarrow$$

($(x-x_k)$ devine $=0$)

$\Rightarrow$ tot timpul $P_n(x_i) = y_i$

Problema examen

Determinați polinomul Lagrange pentru:

x	-1	0	1	2
y	5	1	0	4

$$P_3(x) = \sum_{i=0}^{3} y_i \prod_{\substack{k=0 \\ k \neq i}}^{3} \frac{x - x_k}{x_i - x_k}$$

$$P_3(x) = 5 \cdot \frac{(x-0)(x-1)(x-2)}{(-1-0)(-1-1)(-1-2)} + 1 \cdot \frac{(x+1)(x-1)(x-2)}{(0+1)(0-1)(0-2)}$$

$$+ 0 \cdot \dots + 4 \cdot \frac{(x+1)(x-0)(x-1)}{(2+1)(2-0)(2-1)}$$

Neajunsuri - Interpolare Lagrange

→ are oscilații foarte mari la capete

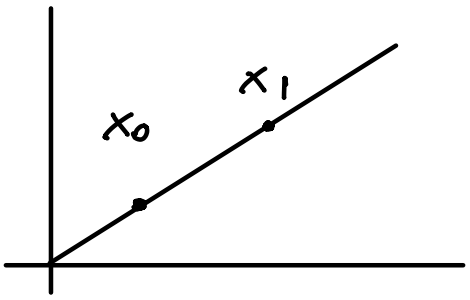
→ erorei foarte mari în determinarea valorilor în punctele intermediare

→ nu putem evalua o eroare globală

→ e bună când sunt puține puncte / pt. punctele din centrul interpolării

(3) Polinomul Newton

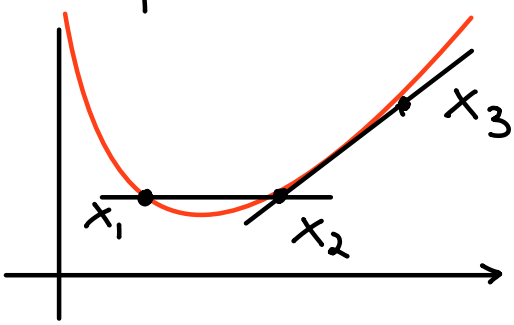
liniară



$$m = \frac{f(x_1)-f(x_0)}{x_1-x_0} = \frac{f(x)-f(x_0)}{x-x_0}$$

$$f(x) = \underbrace{\frac{f(x_1)-f(x_0)}{x_1-x_0}}_{f[x_1, x_0]}(x-x_0) + f(x_0)$$

pătratică



$$\underbrace{\frac{f[x_1\ x_2]-f[x_0\ x_1]}{x_2-x_0}}_{f[x_0\ x_1\ x_2]} = ct =$$

$$= \frac{f[x_1\ x]-f[x_0\ x_1]}{x-x_0}$$

$$f([x_0 x_1]) = (x - x_1) \cdot f[x_0 x_1 x_2] + f[x_0 x_1]$$

$$\frac{f(x) - f(x_1)}{x - x_1} = (x - x_1) f[x_0 x_1 x_2]$$

$$f(x) = (x - x_1)(x - x_0) f[x_0 x_1 x_2] + (x - x_1) f[x_0 x_1] + f(x_1)$$

$$= (x - x_0)(x - x_1) f[x_1 x_0 x_2] + (x - x_1) f[x_1 x_0] + f(x_1)$$

$$f(x) = b_0 + b_1 (x - x_0) + b_2 (x - x_0)(x - x_1) + \dots$$

$$b_k = f[x_0 \dots x_k]$$

$$f[x_0 x_1 x_2 \dots x_{k-1} x_k x_{k+1}] = \frac{f[x_1 \dots x_{k+1}] - f[x_0 \dots x_k]}{x_{k+1} - x_0}$$

$$f[x_1] = x_1$$

pt. $f(x_i) = y_i$ cu $i = 0 : n$

$$F_0[x_i] = y_i$$

$$F_1[x_i, x_{i+1}] = \frac{F_0[x_{i+1}] - F_0[x_i]}{x_{i+1} - x_i}$$

$$F_2[x_i, x_{i+1}, x_{i+2}] = \frac{F_1[x_{i+1}, x_{i+2}] - F_1[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

$$P_n(x) = F_0[x_0] + (x - x_0) F_1[x_0, x_1] +$$
$$+ (x - x_0)(x - x_1) F_2[x_0, x_1, x_2] + \dots +$$
$$+ (x - x_0) \dots (x - x_{n-1}) F_n[x_0, x_1 \dots x_n]$$

$$R_n(x) = \left(\prod_{k=0}^{n}(x-x_k)\right) \cdot F_{n+1}[x_0, \dots x_n, x]$$

$$E(x) \le |R_n(x)| \le \left|\prod_{k=0}^{n}(x-x_k) \cdot \frac{f^{(n+2)}(x)}{(n+2)!}\right| \le M$$

Problema examen

Determinati polinomul Newton pentru:

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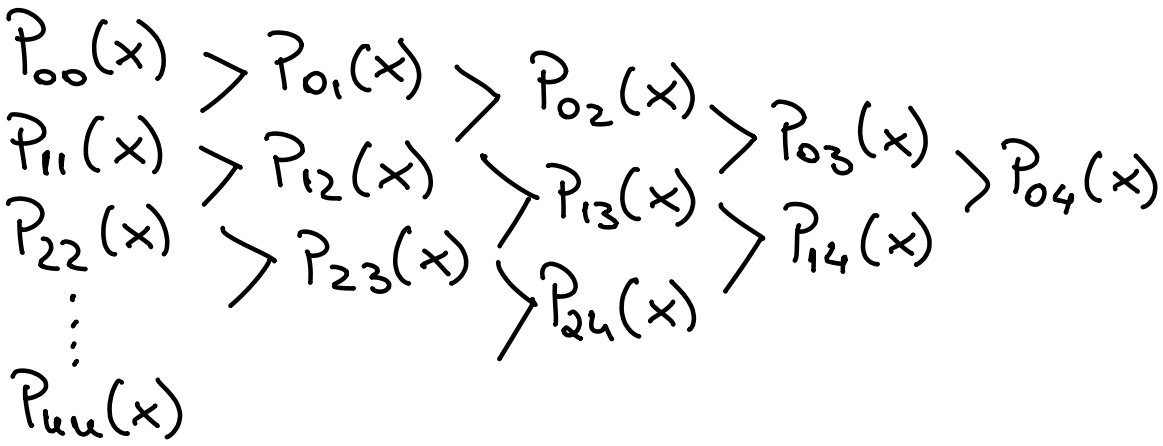
	F_0	F_1	F_2	F_3
-1	5	$\frac{1-5}{0-(-1)} = -4$		
0	1	$\frac{0-1}{1-0} = -1$	$\frac{-1-(-4)}{1-(-1)} = \frac{3}{2}$	$\frac{\frac{5}{2}-\frac{3}{2}}{2-(-1)} = \frac{1}{3}$
1	0	$\frac{4-0}{2-1} = 4$	$\frac{4-(-1)}{2-0} = \frac{5}{2}$	
2	4			

$$P_3(x) = 5 - 4(x+1) + \frac{3}{2}x(x+1) + \frac{1}{3}x(x+1)(x-1)$$

(4) Neville

$$\left.\begin{matrix}(x_1, y_1) \\ \downarrow x \\ (x_n, y_n)\end{matrix}\right\} \Rightarrow P(x)$$

Dintr-un set de date și o valoare x se obține un polinom $P(x) = y$



$$P_{ii}(x) = y_i$$

$$P_{ij}(x) = \frac{(x-x_j)P_{i\,j-1}(x) - (x-x_i)P_{i-1\,j}(x)}{x_i - x_j}$$